Video-CoE: Reinforcing Video Event Prediction via Chain of Events

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Research Questions

- Why do current multimodal large language models (MLLMs) perform poorly on video event prediction (VEP) tasks compared to standard video understanding tasks?
- How can we design a paradigm that enables MLLMs to effectively construct temporal event chains and reason logically about future events from observed video content?
- What training protocols can enhance MLLMs' visual information utilization and temporal reasoning for video event prediction?

Core Idea

Current MLLMs struggle with video event prediction because they rely heavily on textual cues rather than visual evidence and lack logical reasoning capabilities for predicting unseen future events. The key insight is to force models to construct explicit temporal event chains from videos, which acts as scaffolding for both visual grounding and logical reasoning. The Chain of Events (CoE) paradigm segments videos into timestamped event descriptions, then uses these chains as intermediate representations to guide future event prediction, effectively bridging the gap between observed visual content and logical temporal reasoning.

Key Formula

CoE Reward Function

$$r_e^{(i)} = \lambda I(o_i) + (1-\lambda)[L - |\mathrm{len}(o_i) - L| + b]$$

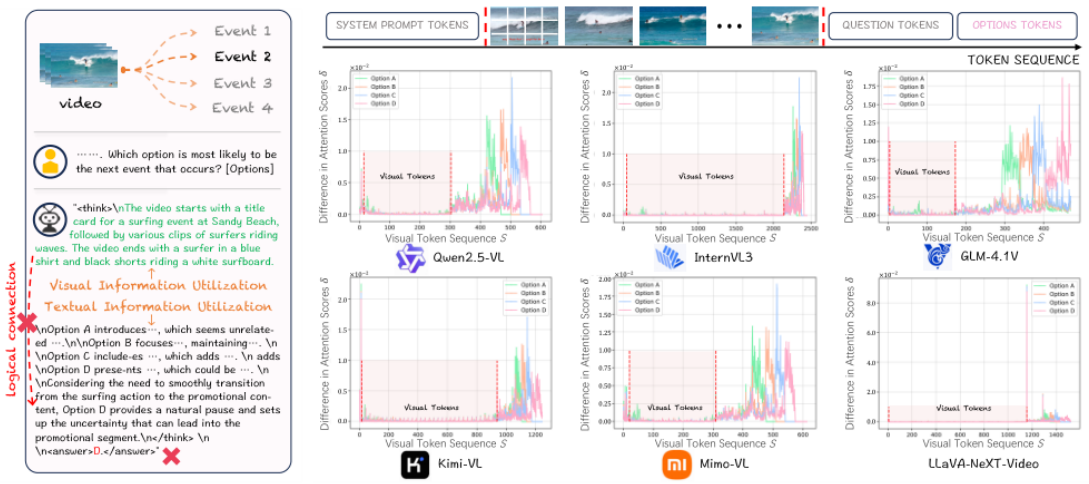
λ is weight coefficient, I(o_i) is indicator function (1 if output contains required event tags, 0 otherwise), len(o_i) counts events in output chain, L is target chain length, b is bias term. This reward encourages both correct event tag construction and optimal chain length for temporal modeling.

Similarity Reward

$$r_s = \frac{1}{n}\sum_{j=1}^{n} s_j, \quad s_j = \cos(v_j, t_j)$$

n is number of events in chain, s_j is cosine similarity between visual feature v_j of video clip j and textual feature t_j of corresponding event description. This reward ensures consistency between event descriptions and actual video content, preventing reward hacking.

Key Figure



(a) An example of MLLMs performing VEP. (b) Attention distribution over input token sequences.

Figure 1. Analysis of open-source MLLMs on video event prediction tasks. Fig. 1a illustrates the reasoning process, indicating the lack of logical reasoning capabilities in VEP task. Fig. 1b illustrates the attention distribution of the option tokens over input tokens demonstrating the insufficient utilization of visual information.

Figure 1: Demonstrates the core problem through attention analysis showing MLLMs allocate substantially less attention to visual tokens than textual tokens during video event prediction, revealing insufficient visual information utilization.

Limitations

- Requires models to have reasonable temporal localization capabilities, which current MLLMs still struggle with
- Event chain construction quality depends on the base model's video understanding capabilities
- Method tested primarily on Qwen2.5-VL; generalization to other MLLM architectures not extensively validated
- Limited to relatively simple event chain structures; more complex temporal relationships (graphs, hierarchies) not explored